Improving Accuracy of Medical Diagnosis Classification

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Introduction

Medical diagnosis classification is traditionally performed by manually coding narrative patient notes into structured data dictionaries, a process that is both error-prone and costly—estimated to account for nearly 12% of healthcare expenses. Automating this step could significantly reduce administrative burden while improving accuracy in medical record systems and research applications. In this study, we tested the effectiveness of fine-tuned large language models (LLMs) compared to baseline models like OpenAI GPT and LLaMA, using the National Electronic Injury Surveillance System (NEISS) dataset. While baseline models achieved only ~66–68% accuracy, fine-tuned models consistently reached ~90–93% accuracy. These results demonstrate the potential of generative fine-tuned models for advancing medical coding, ontology development, and billing systems.

12%
Healthcare Costs
Attributed to manual medical coding and classification

93%
Accuracy
Achieved by fine-tuned OpenAI models

68%
Accuracy
Base models

Objectives

The main objective of this research was to evaluate whether fine-tuned LLMs could substantially improve the accuracy of medical diagnosis classification from injury narratives. We hypothesized that models trained on domain-specific data would outperform baseline models due to improved contextual understanding of medical terms.

Accuracy Evaluation

Measure classification accuracy using both exact (ROUGE) and semantic (Google USE) metrics against ground truth data.

Model Comparison

Compare performance between baseline and fine-tuned models across OpenAI and LLaMA frameworks.

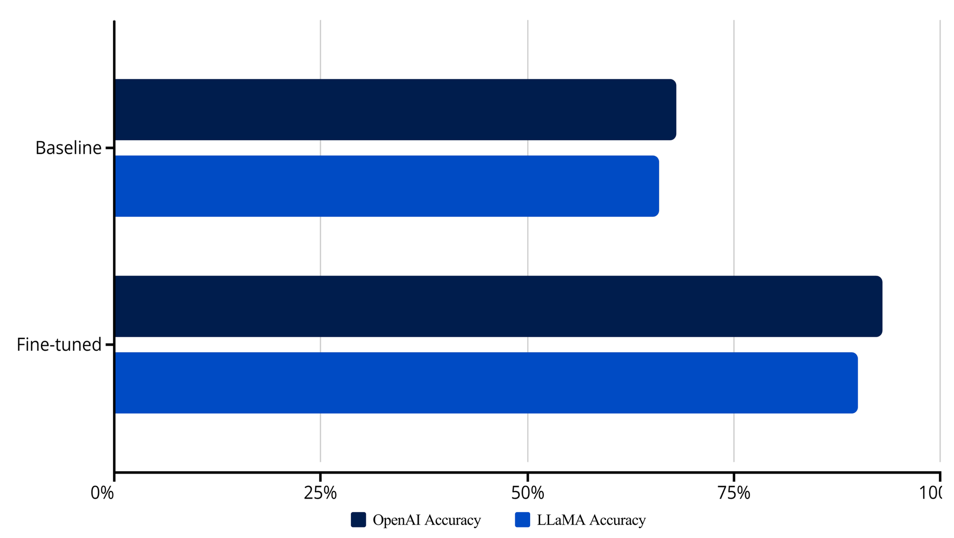
Clinical Viability

Demonstrate potential for billing codes, medical ontologies, and administrative applications by using real world data

Results

The baseline classification accuracy was 68% for OpenAI and 66% for LLaMA. After fine-tuning, OpenAI models achieved 93% accuracy, while fine-tuned LLaMA reached ~90%. Both exact and semantic evaluations confirmed significant improvements, validating our hypothesis. This demonstrates that fine-tuned generative models are highly adaptable for complex medical classification tasks.

Results: Accuracy



Accuracy
Fine-tuned models showed **25-24 percentage point improvements** over baseline accuracy

Validation
Results remained **stable across test and validation sets**, confirming robustness

Performance
Outperformed both baseline LLMs and traditional NLP techniques

Materials and Methods

Dataset and Ground Truth

This study used 20,000 narratives from the National Electronic Injury Surveillance System (NEISS) collected between 2004 and 2024. These narratives describe patient injuries in free-text form and were annotated by researchers at the University of Pennsylvania Orthopedic Lab. The annotations provided structured labels for both the body part affected and the type of injury, drawn from a standardized medical data dictionary. This served as the ground truth for evaluating the classification performance of baseline and fine-tuned models.

20K+
NEISS Narratives
Injury descriptions from 2004-2024 spanning diverse medical cases

20
Years of Data
Temporal span ensuring comprehensive injury description coverage

50
Batch Size
Used batch size of 50 narratives at a time to save inference and training costs

Modeling and Fine-Tuning Approach

Developed three Python programs to process and classify the narratives. To optimize efficiency, narratives were processed in batches of 50, which turned out to be a local optimal batch size for the models. Large batch sizes could work for newer models.

Prompt Engineering
Prompts and narratives as input to convert each narrative into structured injury-body part pairs

Accuracy Evaluation
ROUGE scores for exact matches, Google USE for semantic similarity

Fine-Tuning using expert annotations
OpenAI and TogetherAI APIs enable domain-specific model adaptation

Examples

NEISS Narrative	Expert Classification (Ground Truth)	Predicted (Baseline model)	Predicted (Fine-Tuned)
PLAYING TACKLE FOOTBALL, TOOK "HARD HIT", HAD WIND KNOCKED OUT, THIS AMC/O NECK PAIN, CHEST PAIN, ARM&THIGH PAIN. DX. MUSCLE ACHES	all parts body other/not stated	neck pain	Neck strain
PT GOT HIT IN MOUTH PLAYING FOOTBALL PT SPITTING OUT BLOOD	mouth laceration	no diagnosis found.	mouth laceration
12 Y.O. W MALE W/CONCUSSION. PT INJ WHILE PLAYING FOOTBALL. E9170	head concussions	no diagnosis found.	head concussions
PT PLAYING FOOTBALL IN A CREEK AND GOT SOMETHING DX FB EYE	eyeball foreign body	no diagnosis found.	eyeball foreign body
PLAYING FOOTBALL IN GYM AND RAN INTO ANOTHER KID	face laceration	no diagnosis found.	face laceration
19 YO M PRACTICING FOOTBALL HIT IN NOSELAC NOSE	face laceration	no diagnosis found.	face fracture
SLIPPED PLAYING FOOTBALL YESTERDAY, LBP, HARD TO WALKDX: LBP	lower trunk other/not stated	finger fracture	lower trunk other/not stated
STRUCK IN FLANK AREA AT FOOTBALL PRACTICE;HEMATURIA NOW	lower trunk other/not stated	finger fracture	8: Lower Trunk Internal Organ Injury
PT HIT ON NOSE BY ELBOW WHILE PLAYING FOOTBALL YESTERDAY. D: NASAL CONTUSION.	face contusions, abrasions	no diagnosis found.	9: Face Contusions, Abrasions
PT WITH SUBSTERNAL CHEST PAIN WHICH BEGAN WHILE PLAYING FOOTBALL. D: CHEST WALL PAIN, POSSIBLE GERD.	upper trunk other/not stated	thumb sprain	10: Upper Trunk Other/Not Stated
PT WAS HIT IN HEAD DURING FOOTBALL PRACTICE	head contusions, abrasions	no diagnosis found.	skipped
PATIENT WOKE WITH TESTICULAR PAIN, SWOLLEN, WAS HIT DURING FOOTBALL 2-3WEEKS AGO; TESTICULAR TORSION	pubic region other/not stated	no diagnosis found.	11: Pubic Region Other/Not Stated
PLAYING FOOTBALL FRACTURE RIGHT FOOT	foot fracture	no diagnosis found.	12: Foot Fracture
PT SUST HEAD INJURY WHILE PLAYING FOOTBALL	head internal organ injury	no diagnosis found.	13: Head Internal Organ Injury
PT FELL INTO A TREE WHILE PLAYING FOOTBALL. DX CHI	head internal organ injury	no diagnosis found.	14: Head Internal Organ Injury
15 Y.O. W MALE W/CHI. PT INJ WHILE PLAYING FOOTBALL. E9170	head internal organ injury	no diagnosis found.	15: Head Internal Organ Injury
POSSIBLE BROKEN COLLARBONE - PT INJURED WHILE PLAYING FOOTBALL X4 DAYSAGO. D: NO DX; PT LEFT WITHOUT BEING SEEN BY PHYSICIAN	shoulder fracture	no diagnosis found.	shoulder fracture
PT WAS PLAYING FOOTBALL AND FELL ONTO LEFT SHOULDER. D: LEFT CLAVICLEFRACTURE	shoulder fracture	no diagnosis found.	shoulder fracture

Conclusions

The study confirmed that fine-tuned generative models outperform both baseline LLMs and traditional NLP methods in medical diagnosis classification. Insights from batching strategies and prompt engineering provide useful guidance for future applications of fine-tuned LLMs in healthcare. This approach represents a meaningful step toward integrating AI into medical data infrastructure at scale.

>90% accuracy
90-93% accuracy comparable to human accuracy on medical diagnosis classification

Healthcare Cost Reduction
Reduced manual coding burden could significantly cut healthcare administrative expenses

Clinical Research Potential
Potential for ICD-10, DRG codes, and medical ontology development

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Acknowledgements

I would like to thank Prof. Chamith Rajapakse and Aarush Sahni for their guidance in this project and providing the annotated data to fine tune the models.

Next steps

- Expand testing to 200K NEISS narratives and include clinical features
- Test against other medical diagnosis classification datasets
- Use medically trained models like Google MedGemma to compare accuracy
- Benchmark for ICD-10, DRG and other applications

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